PPE Safety Equipment Detection

Final Technical Summary P14 — Computer Vision 2026

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Problem and dataset

Our objective is to detect personal protective equipment (PPE) and dataset-defined missing-equipment categories, and determine whether YOLO11s improves accuracy enough to justify its cost over YOLO11n. The system outputs bounding boxes, class labels and confidence scores. Assigning equipment to individual workers and determining overall safety compliance are outside the project scope.

We use Ultralytics Construction-PPE: **1,416 images, 11,520 boxes and 11 classes**: helmet, gloves, vest, boots, goggles, Person, none, no_helmet, no_goggle, no_gloves and no_boots. The audit checked image-label pairs, box coordinates, class examples and related scenes. Ten orphan label files were excluded and one duplicate box was removed. Related views and transformed copies were grouped before training, producing a frozen **1,265/73/78 train/validation/test split**. Known reviewed cross-split pairs were eliminated; the audit does not prove that every possible scene relationship was found. The test set contains 924 boxes. Source attribution, the AGPL-3.0 notice and archive checksum are retained.

Methodology and experimental setup

We fine-tuned COCO-pretrained YOLO11n and YOLO11s for **50 epochs with seeds 42, 43 and 44**, using batch 16, 640-pixel inputs, SGD learning rate 0.01, momentum 0.937, weight decay 0.0005, three warm-up epochs and cosine decay. Training uses horizontal flips and mild scale, translation and HSV changes; mosaic, mixup and copy-paste are disabled. Evaluation uses aspect-preserving square letterboxing to 640×640, FP32 and class-aware NMS at IoU 0.70, retaining at most 300 detections. Each seed's checkpoint is selected by validation mAP50:95.

A validation-only confidence sweep maximized mean class macro-F1 across seeds, fixing **0.20 for YOLO11n and 0.30 for YOLO11s before test inference**. All six retained checkpoints are reported; neither thresholds nor seeds were selected on test results. AP uses ranked predictions at confidence 0.001 and the library's multi-label validation. Operating-point precision, recall and F1 use single-label predictions with greedy, one-to-one, same-class matching at IoU ≥ 0.50. Macro-F1 averages class F1 values, not the harmonic mean of macro precision and recall. All 11 test classes have support.

Main results and computational cost

Held-out test metric	YOLO11n	YOLO11s
mAP50:95	0.1874 ± 0.0037	0.1902 ± 0.0067
mAP50	0.4413 ± 0.0150	0.4505 ± 0.0203
Macro precision	0.4731 ± 0.0547	0.5512 ± 0.0307
Macro recall	0.4632 ± 0.0061	0.4525 ± 0.0164
Macro-F1	0.4547 ± 0.0238	0.4835 ± 0.0218
Parameters (unfused)	2.61 M	9.43 M
CPU latency median [IQR], ms	50.21 [46.26, 53.38]	79.80 [68.08, 85.78]

Metrics are mean ± sample SD across three seeds, not confidence intervals. YOLO11s gains **0.0028 mAP50:95 and 0.0288 macro-F1**, with higher precision and slightly lower recall. The small AP difference does not establish statistical significance. CPU timing used one M2 Max, four PyTorch threads, batch one, FP32, seed 42, confidence 0.25, 20 warm-up and 100 timed calls. It includes preprocessing, inference, NMS and call overhead, excluding disk I/O and loading weights. YOLO11s is **1.59 times slower** and has 3.62 times as many parameters. Training took 50.7 ± 0.9 min/seed for YOLO11n (M4/MPS) and 88.9 ± 0.1 for YOLO11s (M2 Max/MPS, three concurrent jobs); these are not comparable training-speed measurements.

Difficult cases and failure analysis

Helmet, vest and Person are strongest: mean AP50 for YOLO11n/YOLO11s is 0.903/0.904, 0.830/0.812 and 0.768/0.782. Goggles is unstable (0.106/0.202; nine test boxes), while no_boots declines from 0.111 to 0.043. More capacity does not consistently improve rare classes.

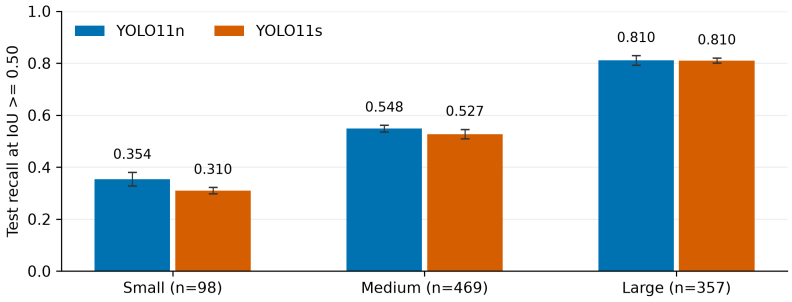


Figure 1. Object-weighted test recall at frozen model thresholds; error bars show sample SD across three seeds. Areas after resize define small $A < 32^2$, medium $32^2 \leq A < 96^2$, large $A \geq 96^2$. Both models miss most small objects. The comparison includes threshold effects and different class mixtures across sizes.

For five PPE classes, clear-object recall is 0.835/0.829 (259 objects), versus 0.676/0.676 for occluded objects (72); 190 uncertain objects are excluded. Labels come from one assistant review before predictions; independent second review is incomplete. This association is confounded by size/class: 58 of 72 occluded objects are vests.

Three seed-42 failures illustrate the remaining problems (Figure 2). In `image182.jpg`, both models miss a 24×28-pixel helmet while detecting larger neighbours, consistent with limited detail and crowding. In `image1037.jpeg`, both miss an occluded vest among overlapping workers; ambiguous annotation boundaries may contribute. In `image1125.jpg`, high-confidence Person boxes span multiple people under umbrellas: IoU is only 0.387/0.330, yielding false positives and a missed person. Confidence filtering alone cannot fix these localization errors.

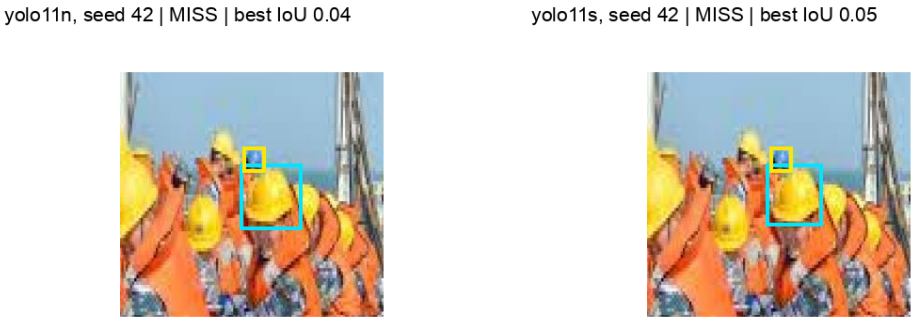


Figure 2(a). Small helmet: image182.jpg, label line 7. All panels compare YOLO11n (left) and YOLO11s (right), seed 42, confidence 0.20/0.30. **Yellow:** selected ground truth; **cyan:** highest-IoU retained same-class prediction. Here, cyan boxes match a larger neighbour, not the selected helmet (IoU 0.045/0.050). Crops are enlarged; inference used full images.



Figure 2(b). Occluded vest: image1037.jpeg, label line 10. The selected vest is missed. Cyan boxes are assigned to a neighbouring worker, with IoU only 0.185/0.159 to this target; they cannot be reused. Overlap, instance separation and ambiguous source box boundaries may contribute.



Figure 2(c). Merged people: image1125.jpg, label line 10. Cyan boxes merge several people (confidence 0.837/0.798; IoU 0.387/0.330). They are unmatched to any ground truth at IoU 0.50, producing false positives; the selected person is also missed. Overlap and merged silhouettes plausibly contribute.

Conclusions and limitations

The larger model offers a modest F1/precision gain at appreciable computational cost. YOLO11n is a reasonable efficiency baseline; YOLO11s is useful when this measured operating-point gain outweighs latency. Neither model adequately handles small objects or all missing-equipment classes. Next steps are more diverse high-resolution data, independent annotation review and validation-controlled small-object experiments.

The small validation/test sets, class imbalance and ambiguous labels limit generalization. Three seeds quantify training variation on one split, not uncertainty over new sites. Visibility analysis is observational and lacks independent human agreement. MPS kernels may remain nondeterministic despite fixed seeds; the reproduction uses CPU loss/NMS workarounds. These findings concern the retained checkpoints; the earlier Stage 3 T4 run, whose weights were lost, is excluded. Performance here does not validate a system for autonomous safety decisions.

Reproducibility and team contributions

The repository contains the frozen split manifest, pinned dependencies, preparation/training/evaluation scripts, six selected checkpoints, per-seed/class results and setup/run instructions. Test evaluation used Python 3.14.7, PyTorch 2.14.1 and Ultralytics 8.4.175 on macOS 15.7.2. The recorded team role allocation is: **Andrei Dubinin** - dataset audit, split preparation and class statistics; **Evgeniy Pestrovskiy** - environment, YOLO11n baseline and run instructions; **Oleg Bondarev** - YOLO11s comparison, thresholds, metrics and timing; **Aleksandr Shiian** - failure analysis, figures and final reporting. Assistant support was used for implementation, analysis and report preparation, including the visibility review disclosed above.

Code and evidence: https://github.com/Ta6leTkaa/intro-to-CV-Project. Metrics and failure evidence: `reports/results_checkpoint`; detailed reports: `planning`.